

Skills Summary

Transformations of Absolute-Value Functions

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading the Vertex and the Transformations

Vertex form: $g(x) = a|x - h| + k$. **Vertex** (h, k) ; **axis of symmetry** $x = h$.

What each letter does to $y = |x|$: h shifts horizontally (right when $x - h$, left when $x + h$); k shifts vertically; $a < 0$ flips the V so it opens down; $|a| > 1$ makes it narrower, $0 < |a| < 1$ makes it wider.

Fast way to find h : set the inside of the bars equal to 0 and solve.

Example: Find the vertex of $g(x) = -2|x + 5| + 3$ and describe the graph.

Step 1. The vertex is the one point where the distance inside the bars is zero, so set the inside to zero and read k off the outside.

Step 2. $x + 5 = 0 \Rightarrow x = -5$; then $g(-5) = -2|-5 + 5| + 3 = 3$. Since $a = -2$, the V opens down and is narrower than $y = |x|$.

$\Rightarrow$ vertex **$(-5, 3)$**

Try it: Find the vertex of $g(x) = 4|x - 6| - 1$.

(1- '9) :pæp

(!) Watch out: $|x + 3|$ shifts LEFT, not right — rewrite it as $|x - (-3)|$ and the sign stops fooling you. And a multiplies only the bars, so in $3|x + 2| - 7$ the 3 never touches the -7 .

Skill 2 — Sketching the Graph from Vertex Form

Three steps: (1) plot the vertex (h, k) ; (2) from the vertex go right 1 and up a to get a point on the right arm, then left 1 and up a for the left arm; (3) draw two straight rays — an absolute-value graph is never curved.

x -intercepts: solve $a|x - h| + k = 0$. They are always the same distance left and right of $x = h$.

Example: Sketch $y = |x - 1| - 4$ and give its x -intercepts.

Step 1. Vertex form is already given, so plot the vertex first and then find where the two arms cross the axis.

Step 2. Vertex: $x - 1 = 0 \Rightarrow x = 1$, and $y = -4$ there. Intercepts: $|x - 1| = 4 \Rightarrow x - 1 = \pm 4$.
 $\Rightarrow$ vertex **$(1, -4)$** , $x = -3$ or **5**

Try it: Give the vertex and both x -intercepts of $y = |x + 2| - 3$.

1 to 9 = x '8- '2-) :pæp

Skill 3 — Solving with the Graph in Mind

Why there are two answers: a horizontal line above the vertex of an upward V cuts it twice, and the two crossings sit the same distance on either side of $x = h$.

Method: isolate the bars, then split. $a|x - h| + k = c \Rightarrow |x - h| = \frac{c - k}{a} \Rightarrow x = h \pm \frac{c - k}{a}$.

Example: Solve $3|x - 2| - 5 = 7$.

Step 1. The bars are not alone yet, so isolate them first; only then can the V be split into its two arms.

Step 2. $3|x - 2| = 12 \Rightarrow |x - 2| = 4 \Rightarrow x = 2 \pm 4$.
 $\Rightarrow x = -2$ or $x = 6$

Try it: Solve $2|x + 1| - 3 = 9$.

check: $x = -2 \Rightarrow 2|-2 + 1| - 3 = 9$

(!) **Watch out:** divide by a BEFORE splitting the bars. Splitting $3|x - 2| = 12$ into $3x - 2 = \pm 12$ loses the bars and both answers.

Skill 4 — Writing the Equation from the Graph

Read the vertex, then one more point. The vertex hands you h and k at a glance. Substitute any second point (x_1, y_1) into $y = a|x - h| + k$ and solve the resulting one-variable equation for a : $a = \frac{y_1 - k}{|x_1 - h|}$.

Sanity check: if the second point is BELOW the vertex, a must come out negative.

Example: A V has vertex $(2, 1)$ and passes through $(5, 7)$. Write its equation.

Step 1. The vertex gives two of the three unknowns for free, so the extra point is only needed to pin down the steepness a .

Step 2. $h = 2, k = 1$, so $7 = a|5 - 2| + 1 \Rightarrow 3a + 1 = 7 \Rightarrow a = 2$.
 $\Rightarrow y = 2|x - 2| + 1$

Try it: A V has vertex $(-4, 6)$ and passes through $(-1, 0)$. Write its equation.

check: $y = 2|x + 4| + 6$